

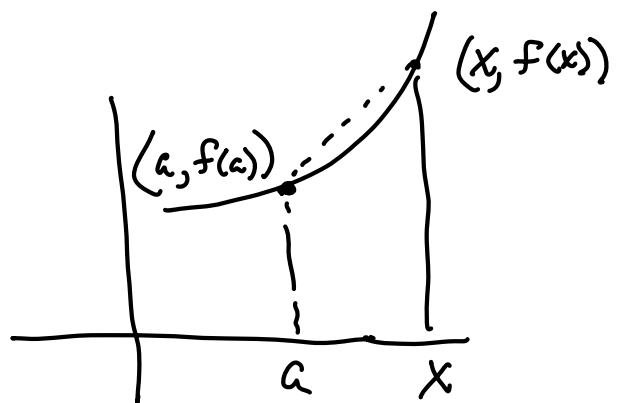
Class 31 Derivatives

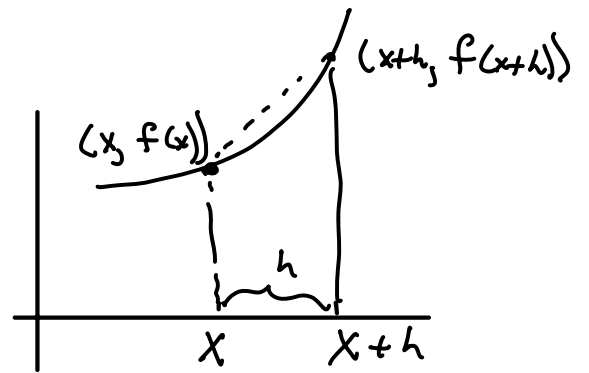
Goals

1. Know both forms for the definition of the derivative.
2. Be able to use the definition of derivative and limit rules to find the derivative of a given function.
3. Know the connection between continuity and differentiability.

What does the derivative tell us about a function, f ?

To determine the instantaneous rate of change at $x = a$ or the slope of the tangent line at $(a, f(a))$, we need to get a sequence of average rates or slopes over an interval of x -values and take the limit of these averages as the size of the interval goes to 0. If this limit is a number, we say the derivative exists and is equal to that number.





Rules to calculate derivatives arise from rules for limits.

Sum Rule $D_x[f(x) + g(x)] = f'(x) + g'(x)$

Proof:

Product Rule $D_x [f(x) \cdot g(x)] = g(x) f'(x) + f(x) \cdot g'(x)$

Proof:

Theorem: If f is differentiable at $x = a$, then f is continuous at $x = a$.

Proof: